

Introduction to Modeling Linear Relationships

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Agenda

- Overview of Machine Learning
- Simple Linear Regression
- Multiple Linear Regression
- Handling Categorical Variables
- Evaluation Metrics for Regression

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Let's begin the discussion by answering a few questions

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Linear Regression Quiz

Which of the following tasks is typically associated with regression problems?

A

Predicting whether an email is spam or not

B

Segmenting customers into different clusters based on their purchasing behavior

C

Predicting the price of a house based on its features

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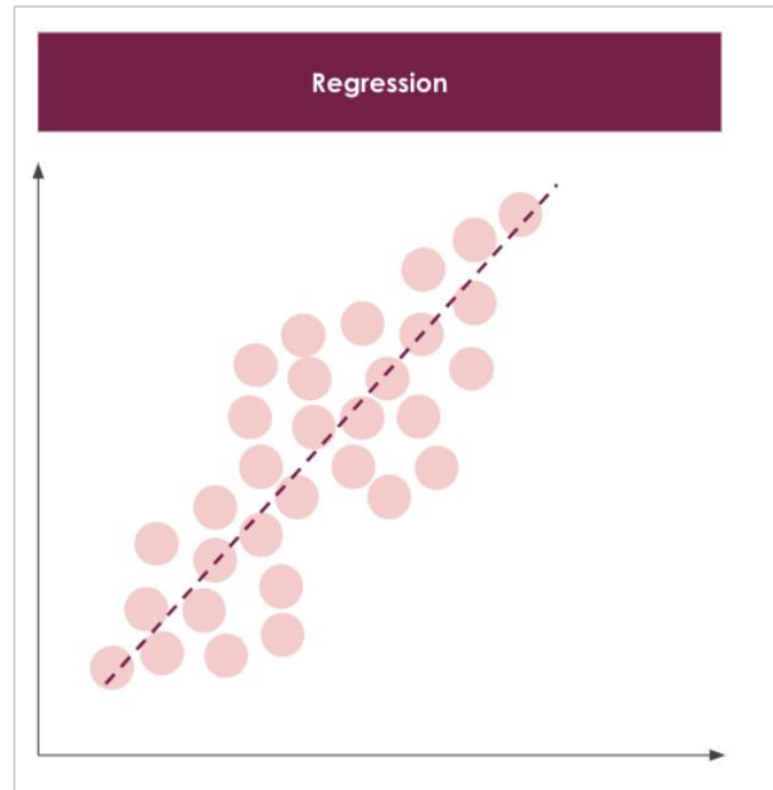
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Simple Linear Regression

Regression is a type of **supervised learning task**, which focuses on **predicting a continuous numerical value or quantity** with the given input features.

This can be thought of more as **drawing a "trend line" in vector space**

Predicting the price of a house based on its features (such as size, number of bedrooms, location, etc.), is a classic example of a regression task. The goal is to predict a continuous value (the price) based on various features.



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Linear Regression Quiz

In a linear regression model,

$$\text{house_price} = 50,000 + 200 * \text{area(sq ft)} + 150 * \text{number_of_rooms}$$

For a 1200 sq ft house with 3 rooms, what will be the price?

A

\$ 50,450

B

\$ 290,000

C

\$ 290,450

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Multiple Linear Regression

To find the value of the **dependent variable (house_price)** for the given test data point, we substitute the values of the **independent variables (area = 1200 sq ft and number_of_rooms = 3)** into the **regression equation**.

$$\text{house_price} = 50,000 + 200 * (\text{area sq ft}) + 150 * (\text{number of rooms})$$

$$\text{house_price} = 50,000 + 200 * 1200 + 150 * 3$$

$$\text{house_price} = 50,000 + 240,000 + 450$$

$$\text{house_price} = 290,450$$

So, the predicted house price for the given test data point is \$290,450.

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Multiple Linear Regression

Dependent Variable

The variable that is being predicted or explained in the regression model

Independent Variables

The input variables that influence the outcome of the independent variable

$$y = b_0 + b_1x_1 + b_2x_2 + \dots$$

Intercept

The constant term that represents the value of the dependent variable when all independent variables are zero

Coefficients

The weights assigned to each independent variable, indicating their impact on the dependent variable

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Linear Regression Quiz

Given a linear regression equation:

$$y = b_0 + b_1x_1 + b_2x_2$$

If x_2 increases by a value of 2, what will be the corresponding increase in the value of y ?

A

$$2b_0$$

B

$$2b_1$$

C

$$2b_2$$

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Multiple Linear Regression Interpretation

From the original equation:

$$y = b_0 + b_1x_1 + b_2x_2$$

We can see that the dependent variable(y) is related to the independent variable x_2 by a factor of b_2 .

This means that for every unit change in x_2 , there will be a corresponding b_2 change in the dependant variable y .

So, if x_2 increases by **2 units**, y will increase by **$2b_2$ units**.

Note: This interpretation holds true ONLY when the assumptions of linear regression are satisfied.

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Linear Regression Quiz

Why can't categorical variables be directly used in linear regression?

A

They have a small variance.

B

They add little value to the prediction power of a model.

C

They represent categories while the best fit line needs to be fit on numerical values.

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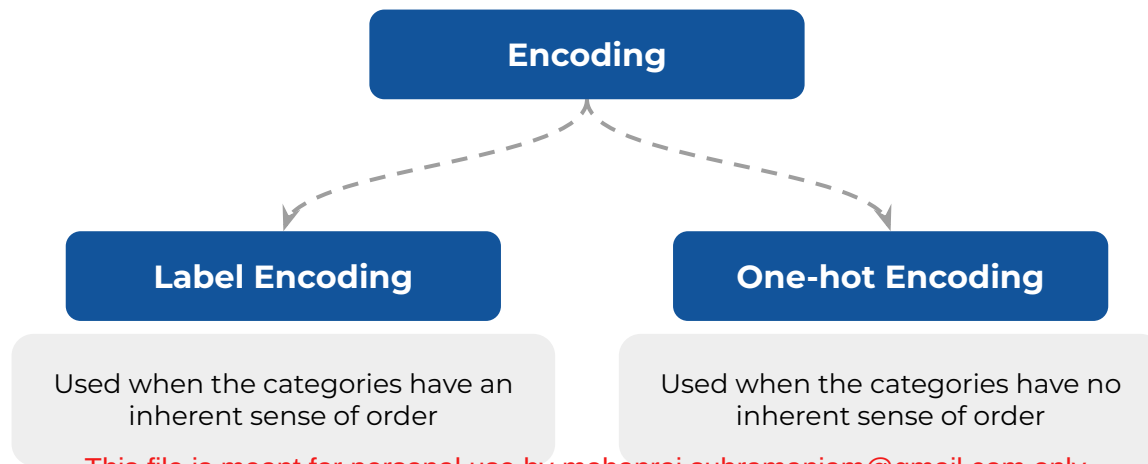
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Encoding of Categorical Variables

Categorical variables are not numbers - even if they might be represented by numbers

Can't be utilized directly in a linear regression model

Need to be converted into a numerical format



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Linear Regression Quiz

Which of the following metrics measures the proportion of total variance explained by the model?

A

Mean Absolute Error(MAE)

B

Mean Squared Error (MSE) / Root Mean Squared Error (RMSE)

C

R-Squared (R^2)

D

Adjusted R-Squared (Adj. R^2)

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Evaluation Metrics in Regression

R-squared	Adjusted R-squared	Mean Absolute Error	Root Mean Square Error
- Measure of the % of variance in the target variable explained by the model - Generally the first metric to look at for linear regression model performance - Higher the better	- Conceptually, very similar to R-squared but penalizes for the addition of too many variables - Generally used when you have too many variables as adding more variables always increases R^2 but not Adjusted R^2 - Higher the better	- Simplest metric to check prediction accuracy - Same unit as the dependent variable - Not sensitive to outliers i.e. errors doesn't increase too much if there are outliers - Difficult to optimize from a mathematical point of view (pure maths logic) - Lower the better	- Another metric to measure the accuracy of prediction - Same unit as the dependent variable - Sensitive to outliers - errors will be magnified due to the square function - But has other mathematical advantages that will be covered later - Lower the better

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Power Ahead !



Appendix

Metric	Formula
R-squared	$R^2 = 1 - \frac{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2}$
Adjusted R-squared	$Adj. R^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - k - 1} \right]$
Mean Absolute Error	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $
Root Mean Square Error	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$

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